

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Note : This procedure **only** applies to the automotive with CM871 and automotive/industrial CM570 with two piece water pump designs.

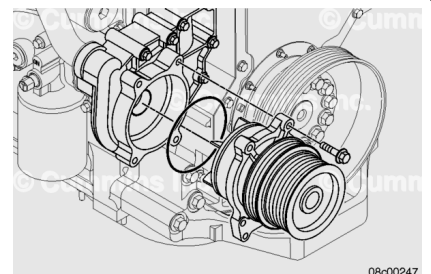
- Drain the cooling system. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>).
- Remove the water pump drive belt and tensioner. Refer to Procedure 008-080 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-080-tr.html>).
- Remove the refrigerant compressor drive belt. Refer to Procedure 009-052 in Section 9 (</qs3/pubsys2/xml/en/procedures/10/10-009-052-tr.html>).

Remove

Remove the water pump cartridge mounting capscrews.

Remove the water pump cartridge and o-ring.

Discard the o-ring.



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Clean and Inspect for Reuse

Clean the water pump cartridge.

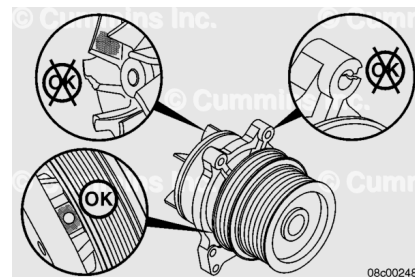
Inspect the water pump cartridge for cracks.

Inspect the water pump weep hole for indication of leaks.

Stains around the weep hole are normal and will be expected and are **not** considered a warrantable malfunction. A warrantable water pump malfunction is defined as stains around the weep hole which form a drip or accumulation of liquid.

Inspect the cast iron impeller for cracks, pitting, and other damage.

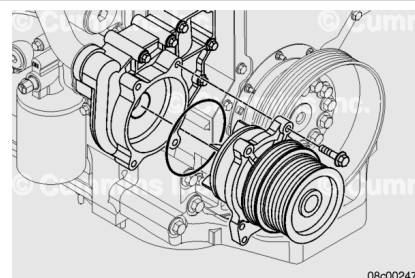
If the indication of a steady flow of coolant is observed, replace the water pump cartridge with a new or rebuilt unit.



Install

⚠ CAUTION ⚠

Do not use petroleum based lubricant. Petroleum based lubricants will cause the o-ring to crack or swell, leading to a coolant leak.



Install the new o-ring seal on the cartridge.

Install the cartridge in the water pump housing.

Install the water pump cartridge capscrews

Tighten the capscrews.

Torque Value:

Four Capscrews 48 n•m [36 ft-lb]

Torque Value:

Six Capscrews 23 n•m [204 in-lb]

Finishing Steps

- Install the refrigerant compressor drive belt. Refer to Procedure 009-052 in Section 9 (</qs3/pubsys2/xml/en/procedures/10/10-009-052-tr.html>).
- Install the water pump drive belt and tensioner. Refer to Procedure 008-080 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-080-tr.html>).
- Fill the cooling system. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>).
- Operate the engine. Check for leaks.

Last Modified: 13-Sep-2018
